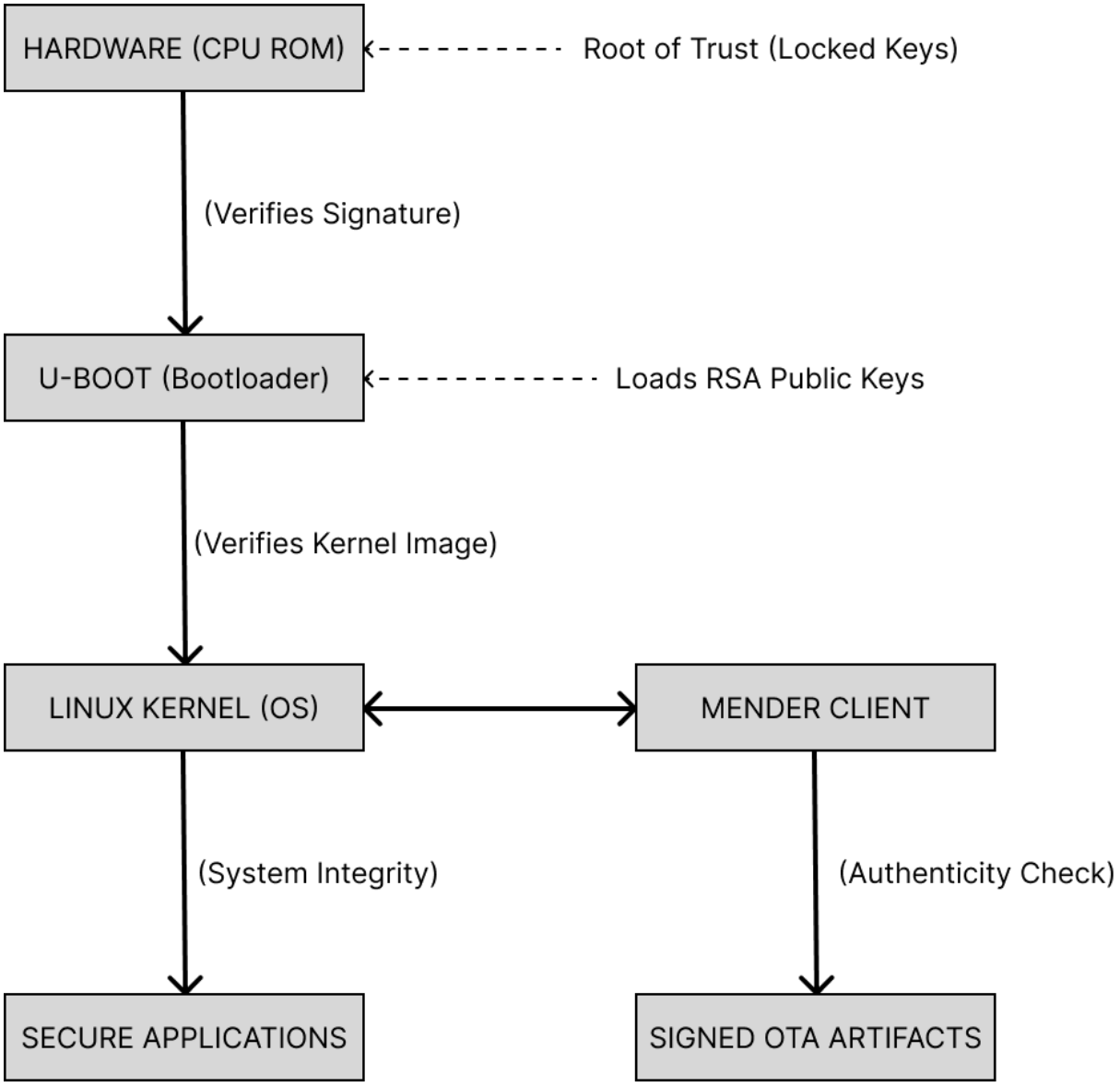
Deliverable 1: Architecture & Process Documentation (Task 1)

1. Objective

To design and implement a high-integrity "Chain of Trust" that ensures only authenticated, signed firmware can execute on the IoT device.

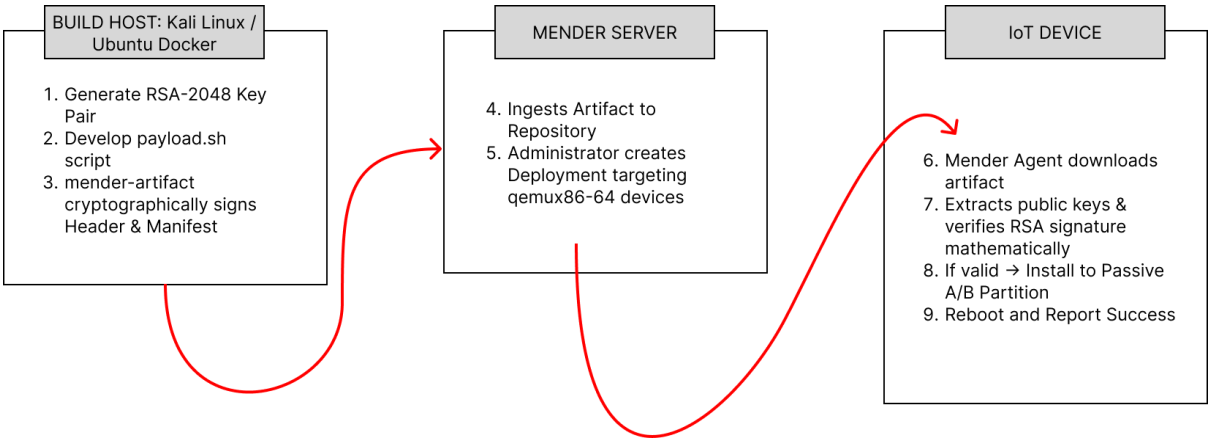
2. Secure Boot Chain

This diagram illustrates the hardware-to-software handoff. Each stage must cryptographically verify the next before execution begins.



3. Firmware Update Verification Flow

This tracks the specific lifecycle of the `custom-signed-v1.0` artifact you successfully deployed.



4. Technical Specifications

- **Architecture:** `qemux86-64`.
- **Bootloader Recommendation:** U-Boot with Verified Boot enabled to support RSA-2048.
- **Partition Strategy:** A/B Redundancy to ensure a "Known Good" state is always available for recovery.